

# Expressive Skin

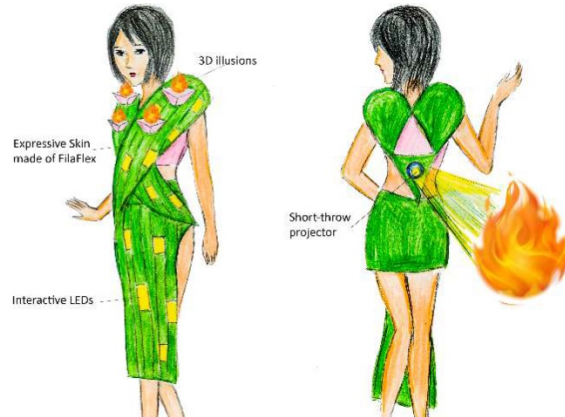
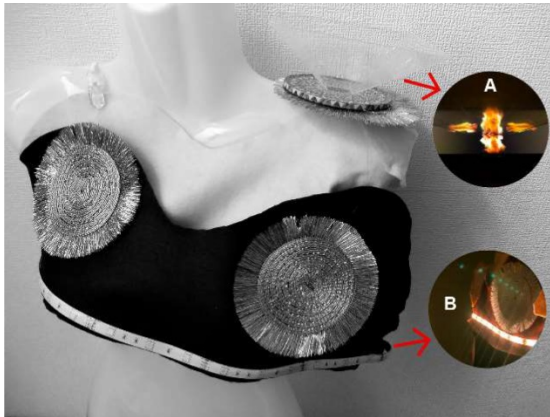
An extended prosthesis that reflects content from the virtual world onto the user's original body

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**Figure 1: Overview of Expressive Skin's low-fidelity prototype (Left) with pyramid hologram projection (A) and color-changing LED lights (B). Full sketch of high-fidelity prototype (Right).**

## ABSTRACT

In this paper, we introduce Expressive Skin, a 3D-printed, haptic display interface that links the virtual world with the real world through the user's body. This research's vision is to imagine an advanced version of the human body with an extended prosthesis that can leverage human cognitive abilities. We believe that by immersing our bodies in virtual content, we foster creative self-expression, imagination, and thus design new lifestyles. This is the concept of Extensive Immersion. The Express Skin is equipped with a projector on its back, four pyramid-shaped projection modules on its front, and a LED lights system strengthening the visualization of gaming content by reacting simultaneously with the projections and with the tactile interaction (Figure 1 Full sketch of high-fidelity prototype). Through the creation of Expressive Skin, we want to ignite the fuse of fashion and gaming, as a viable way to achieve Extensive Immersion.

## CCS CONCEPTS

• Human-centered computing; • Haptic devices.;

## KEYWORDS

Extensive Immersion, Fashion, Game, Haptic Display

## ACM Reference Format:

Chu Yen Ly and Harry Krekouiokitis. 2021. Expressive Skin: An extended prosthesis that reflects content from the virtual world onto the user's original body. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451848>

## 1 INTRODUCTION

Throughout the evolutionary history of video game, the idea of games as a form of self-expression extended beyond the game world itself. From customizing game character's appearance and play style to designing desired lives of their own in open-world games, something they may not be able to do in real life. This endless desire for "making myself heard" in gaming is the origin of various cultural and business trends. Cosplay culture from Japan, digital clothes for sales for game characters, or game-branded clothing collections co-produced by game companies and fashion houses are good examples of how the interest of the audience is captured, maintained, and developed in ways that push the industry forward and blur the boundaries between digital entertainment and other domains in our daily lives. In the future, gaming activity will not merely serve the purpose of self-entertaining. The gaming experience should augment to the extent to which players can recreate themselves and their worlds. How can we facilitate this trend from the designer's point of view? We are finding the answer through the design of Expressive Skin.

**Survey.** To examine the importance of this research, we proposed the definition of Extensive Immersion in Video Games as the state of being empowered to engage in creative activities after playing game, and conducted a survey to find the key determinants of

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*CHI '21 Extended Abstracts*, May 08–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451848>

**Table 1: Survey Result & Scores**

| Game                          | Score | Extensive Immersion Trigger            | Post-game Actions                                                                    |
|-------------------------------|-------|----------------------------------------|--------------------------------------------------------------------------------------|
| Assassin's Creed: Origins     | 4.57  | Realism: Ultra Realistic Game World    | Research on how the game was made, digital 3D animation, ancient Egypt civilization. |
| Final Fantasy VII The Remake  | 3.98  | Storytelling                           | Subscribe to Final Fantasy social media, do some fan arts.                           |
| Samurai Warrior: State of War | 3.54  | Possibility for Making Personal Legacy | Read about the Age of the Samurai, join a Cosplay community.                         |

Extensive Immersion. The survey was conducted both online and offline, involved 40 volunteers in an observable 1-hour game playing session where they were required to choose one game to play from a list of games made by the survey's owner (See Table 1). After completing the session, through a questionnaire, volunteers graded their experiences about the games they chose to play, from 1 point (worst experience) to 5 points (excellent experience) for each of the criteria in Concept of Richness [Werner Wirth et al. 2007]. This Concept of Richness is explained shortly as below:

- (1) *Multiple channels of sensory information*: The more senses we assault and the more those senses work in tandem, the better.
- (2) *Completeness of sensory information*: The fewer blanks about the mental model of the game world that the player has to fill in such as abstractions and contrivances, the better.
- (3) *Cognitively demanding environments*: A world full of challenges that require players to concentrate their brain power.
- (4) *A strong and interesting narrative, plot, or story* that relates to players' personal experiences, opinions and feelings.

Next, they were requested to comment their state of immersion and which aspects of the games made they felt that way. The last one was an optional task, which required them to share the game-inspired, post-game actions they took. The result of the above study pointed out Realism (the quality of in-game content that is accurate and true to life) and Personal Legacy (the possibility of players creating personal imprint in the game experience) are two of the most important factors determining Extensive Immersion in gaming, from which the concept of Expressive Skin was derived. (Table 1)

*Similar Works*. There are several similar researches that inspires us to explore the topic of haptic wearable devices. Hashizume [Hashizume, et al. 2018] designed LIVEJACKET, a wearable integrated with multiple speakers that can surround user with live music. Akira Ishii [Akira Ishii, et al. 2019] with ReverseCAVE opened the VR world – which can be individually experienced through a head mounted display (HMD) – to observers. Noticeably, Synesthesia Wear [Taichi Furukawa, et al. 2019], a full-body haptic clothing interface based on two-dimensional signal transmission can allow users to walk freely in the virtual space and customize the haptic feedback to their own liking.

With Expressive Skin, we want to contribute an alternative solution of immersion enhancement that settles a new game play style, at which gaming meets fashion.

## 2 INITIAL DESIGN: LOW-FIDELITY PROTOTYPE

*Preliminary Test*. To test the idea, we sketched some user scenarios described in Figure 2 and conducted several interviews with target users – professional gamers. As we received supportive feedbacks and suggestions for implementation, according to which we created a low-fidelity prototype of Expressive Skin as shown in Figure 1. This prototype is made of chemical cotton cloth with a simple projection module and LED lights module. The projection module consists of a mini computer interface displaying four symmetrically opposite variations of the same image and a hologram pyramid. In principle, each side projects the image reflecting on it to the center of the pyramid, which creates a 3D illusion. This 3D illusion is the visualization of in-game content on the user's body through the Expressive Skin. The LED lights are designed to react according to the tones, elements, or shades of the displayed visual. Figure 3 describes how the prototype system works.

We held two sets of users test with two different participant groups: observers – non-gamers (7 people) and users – professional gamers (2 people). The user tests took place in X-House share house based in Inukura, Kawasaki, Kanagawa. The goal was to understand different expectations for a game-support wearable like Express Skin from a diverse group of users. Participants wearing the prototype were requested to focus on a game scene of a game character in the fire on the computer screen while the symmetrical images were activated to produce the 3D illusions of flame. At the same time, the LED lights turn red to match with the color of fire (manually controlled).

During the first test, we observed that when a user was wearing the prototype and doing some poses in front of the fire scene and with the illusion of flame on his shoulder, some observers started the conversation about the gaming experience, some kept complimenting the user's cool look and some played with the projections and the LED lights. The user explained that he had the feeling of his being and the game character's being switched with each other for a very short moment. He wished that it would last longer (Figure 4). In the second test, we observed that the same actions were taken by the observers but the interaction was not much. Besides, the user commented that she didn't feel anything special and the reflected content was not overwhelming enough to bring any immersive feeling.

The results might be dependent on various factors, such as users' expectations, preferences, and sensitiveness of surroundings. However, they gave us some takeaway points from which we formed



Figure 2: User Scenarios of Expressive Skin – The interactions Between Game – Player and Player – Observer.

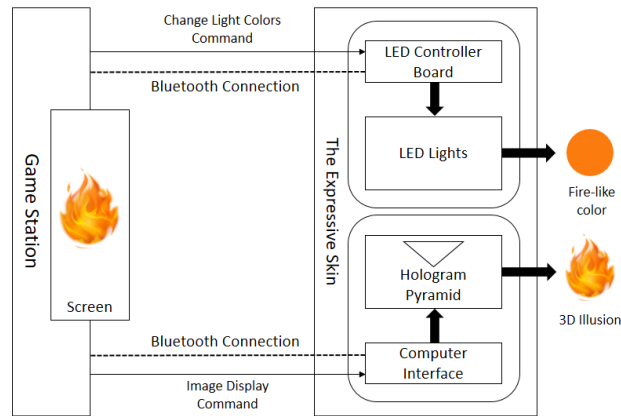


Figure 3: The Expressive Skin's Low-Fidelity Prototype – System Description Diagram

the idea of how to build the next high-fidelity prototype. Thanks to the positive feedback from all participants of the first test, we kept believing that the Express Skin can drive the user to a different dimension of immersion, while ignited the communication with observers as a catalyst for self-expression. From both tests, we learned that the users were not satisfied because the projections and light effects were not immense and interesting enough, and the design of the prototype itself was not yet eye-catching.

### 3 EXPRESSIVE SKIN: HIGH-FIDELITY PROTOTYPE

We have been building the high-fidelity prototype of Express Skin according to the above points of improvements concluded from the first user test (Figure 1).

- (1) *Interactive LEDs*: Users or observers can change the visual effects of the LED lights through tactile interaction. This

effort is to make the Expressive Skin more reactive to impress people.

- (2) *Fashionable Design*: The Expressive Skin should have a more impressive design to facilitate user's self-expression effort. The coverage of projections and LED lights should be larger to create the state of being overwhelmed to users.

This prototype is 3D printed with FilaFlex as the main printing material due to its great elasticity and softness. In addition to the pyramid projections, this prototype is also equipped with a short-throw, laser projector at the back and an interactive LEDs light system at the front. Its futuristic design implies our vision of an advanced human body that enables us to interact with the surrounding environment in novel ways. This is when new communications, new languages, and new cultures are invented. The prototype will be ready by the beginning of February 2021.

#### Users' Quotes

"The Expressive Skin is something that I have always dreamed of as a lifelong gamer. It gave me the feeling that me and Sephiroth (\*) were one."  
Yuutarou Manabe, Student & RPG fan from Japan

"It was amazing to think of my body as an artwork and a point where the unreal meets the real. The Expressive Skin is a cool way to involve my friends into the worlds of my favorite games." Lucas Kikuchi, Student from Brazil

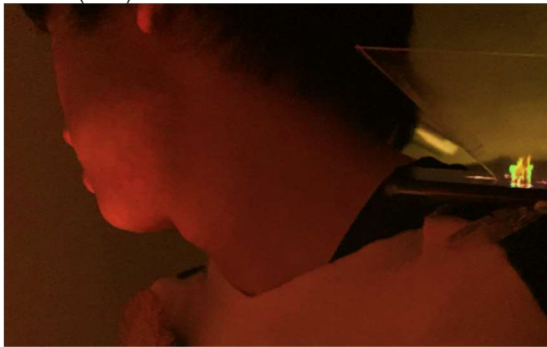
(\*) Sephiroth is the main antagonist of Final Fantasy VII and Final Fantasy VII Remake by Square Enix



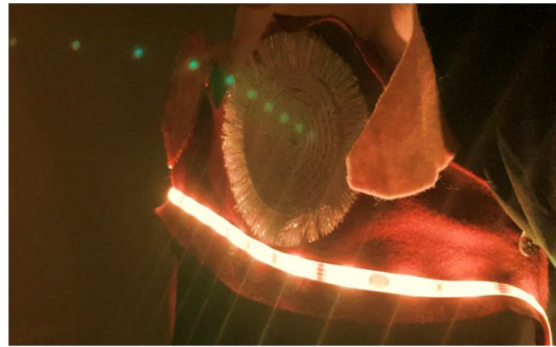
Test Scene: Sephiroth in Nibelheim Incident, Final Fantasy 7 Remake (2020)



Yutarou was focusing on the game scene.



The illusion of fire on his shoulder started to display.



The LED lights happened to change to red orange to match with the color of fire.

Figure 4: Yuutarou Manabe is enjoying the Expressive Skin.

## 4 LIMITATIONS AND FUTURE WORK

*Limitations.* One major limitation of our research was that we were unable to reach a larger number of participants with various demographic identifiers for the preliminary survey and user test. It was also a challenge to synchronize the visualization effects on the prototype with the game scenes.

*Future Work.* In short term, we aim to produce a workable, high-fidelity prototype and participate in more exhibitions to receive more feedback from the audience. In long-term, the Expressive Skin is a part of Extensive Immersion, an umbrella project of which vision is *multiplying the immersion*. At this stage, the Extensive Immersion Project comprises two main subprojects: Express Skin and *The Re-Creation of Sengoku* role-play, AR game. Starting out as a university project, we have the ambition to develop it into a digital entertainment company with the Expressive Skin and the Sengoku-themed game as final products. The development of this game is currently put on hold due to the lack of human and financial resources. The reason for choosing this theme is that the Sengoku period is marked as the most important historical milestone in Japanese long history when Japan performed a massive change in its economy, culture, society, and political system in a challenging turmoil to open the new era ahead – a modern Japan. We believed that there would be a lot of stories to be conveyed through the Express Skin as we travel back to Sengoku Jidai once again. We hope that by taking part in CHI 2021 Game Competition, we will receive some valued feedbacks and different kinds of support from experts and audience so that we can get back to the game development earlier.

## ACKNOWLEDGMENTS

This project was supported by Embodied Media Project, Keio Graduate School of Media Design. We would like to thank Yutarou Manabe for sharing priceless advice regarding how to apply cultural insights into the design process and joining the user test, and Robert Sørleid for gathering up your friends to help us with the Extensive Immersion survey.

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